

OPERATING SYSTEMS

Storage Management

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OPERATING SYSTEMS

Linux File Types

- Regular Files
- Directory Files
- Device Files
- Link Files
- Pipe File



Checking the file type

Command: **ls -l**

- - regular file
- d directory file
- b block device file
- c character device file
- p pipe file
- l link file

Regular File

- It may be text, binary, executable program, program input or output file
- They can be read or written by users with appropriate permissions
- Created using touch, vi, cat or gedit command
- To remove regular files use rm command

Directory File

- Folder that contains other files and subdirectories
- Provides a means to organize files in hierarchical structure
- To create : **mkdir** command
- To remove : **rmdir** command

Device File

- **Block Device File:** Physical device which transmits data a block at a time
- Eg: hard disk drive, floppy disk drive
- **Character device file** : Physical device which transmits data in a character-based manner
- Eg: line printers, modems, consoles
- To create device file: mknod command

Eg: `mknod /dev/sda7 b 45 0`

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`/dev/sda7` : name of the device

`c` -- character device `b` -- block device

`45` — major device number

`0` — minor device number

Major device number : an index to the kernel's file table that contains address of all device drivers

Minor device number : indicates the type of physical file and I/O buffering scheme used for data transfer

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Linux File Types

Floppy disk - /dev/fd0

Tape devices - /dev/rmt0

CD-ROM devices - /dev/cd0

Mouse interface - /dev/mouse

Null devices - /dev/null

Terminal - /dev/tty

Link File

- Creates reference for the file
- Two types of Links can be created
 - **Hard Link** (no option is specified)
 - **Soft/Symbolic link** (-s option is specified)
- To create : **ln** command with **-s** option

Eg: **ln -s** /home/mozhi/dir1/original
/home/mozhi/dir1/slink

Pipe File

- Creates a named pipe file
- It is used only in programs to act like a buffer and used only for interprocess communication
- This file will not contain any contents after the execution of the program
- To create : **mkfifo** command

Eg: **mkfifo** pipe1



THANK YOU

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